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2. Write the python program to solve 8-Queen problem

Aim

To write a pythonw program to solve the 8-Queen problem using the Backtracking algorithm.

Algorithm

1. Start with an empty chessboard
2. place one queen in each row.
3. Check whether the queen conflicts with any previously placed queen.
4. If the position is safe, place the queen and move to the next row.
5. If no safe position is available, remove the previous queen and try another position.
6. Continue until all eight queens are placed safely
7. Display the solution.

Program

```
def Solve (row, queens):
    If row == 8:
        print ("Solution:")
        for r in range (8):
            Print (" ".join(
                "Q" if queens[r] == c else ".":
                for c in range(8)
))
```

```
            return True
        for col in range(8):
            safe = True
            for r in range(row):
                if (queens[r] == col or
                    abs(queens[r] - col) == abs(r - row)):
                    safe = False
                    break
            if safe:
                queens[row] = col
                if solve(row + 1, queens):
                    return True
        return False

queens = [-1] * 8
if not solve(0, queens):
    print("No solution found.")
```

Sample Input

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Sample Output

```
Q . . . . . .
. . . . Q . .
. . . . . Q . Q
. . . . . Q . .
. . Q . . . . .
. Q . . . . . .
. . . Q . . . .
```

Result

Thus, the 8-Queen problem was solved successfully using the Backtracking algorithm